Supplement, 19 October: when a fitting hypothesis is not enough

CORA reasoning v2, a development update for the summit poster. It is not part of any Kaggle entry, and nothing here is certified.

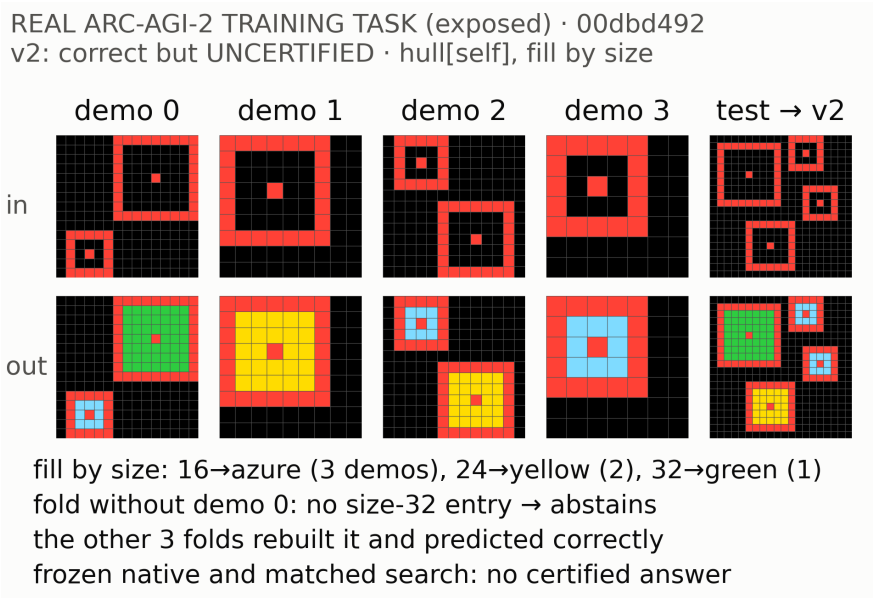
What v2 does.

- A hypothesis is a triple: an object representation (one of CORA's seven segmentations), a geometry (one of 37,750 growth expressions) and a colour policy.
- v2 keeps *every* hypothesis that reproduces every demonstration exactly, predicts the test with each, and abstains when they disagree.
- It reruns the whole search with each demonstration held out, and certifies only a unique prediction that every fold rebuilds. The verifier is CORA's unchanged leave-one-out discipline.

Typed diagnoses. When the hypotheses are not enough, v2 records:

- the symptom (8 labels);
- the competing causes;
- a bounded probe on the demonstrations, and its result;
- what stays uncertain, and the next action.

It reads no test answer. *No program found* is never reported as *new semantics needed*.



The first correct but uncertified prediction (00dbd492, an exposed training task). v2 fills each frame with a colour set by its size. Its 1,365 consistent hypotheses all agree, and the prediction is correct. Frozen CORA and matched extra search give no certified answer.

- **Why certification fails:** the test needs the size-32 colour, which only one demonstration shows, so the fold without that demonstration abstains.
- **What the diagnosis rules out:** the probes reject ambiguity, a representation error and a defect. The cause is insufficient evidence.
- **What it does not change:** a failure-blind reasoner with the same language gives the same answer. This is candidate generation; it does not show that diagnosis helps.

Engineering runs (exposed tasks, in-sample).

- **58 pool tasks:** 0 certified; 3 correct and 1 wrong exploratory predictions. Added to native's recorded Kaggle-path attempts, they would raise the two-attempt count from 7 to 8 (filling only duplicate attempt-2 slots) or to 9 (always as attempt 2), with none lost.
- **60 development tasks:** 0 certified; 1 correct exploratory prediction, on a task native already solves (12 stays 12).
- **Main limitation:** 32 of the 60 change cells in ways v2 cannot express (several actions, or recolouring or deletion only). That is the next hypothesis class to build.

Not shown.

- No certified v2 answer.
- No result on tasks outside development.
- No evidence that diagnosis-driven revision beats failure-blind revision.

Next. A frozen real-task test: v2, frozen CORA and matched search, then revision chosen by the authentic diagnosis, failure-blind and shuffled. It waits for a cohort check, because Step B's protected splits are drawn from the same training file.

Sources: `records/CORA_REASONING_V2_*_20261019.md`; ledger rows 18–21.